Devansh **Shringi**

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Education

University of Toronto Sept. 2022 - Present
Ph.D. in Computer Science; **expected August 2027**.
Supervisor: Shubhangi Saraf.

Indian Institute of Technology Kanpur July 2017 - May 2022
Dual B.Tech.-M.Tech. in Computer Science and Engineering.
Master's supervisor: Nitin Saxena.

Research Interests

Algebraic complexity and algorithms, computational complexity theory, pseudorandomness, and computational algebra.

Publications and Preprints

Fast Algorithms for Low Rank Tensor Decomposition
Shubhangi Saraf, **Devansh Shringi**
Submitted manuscript, 2026 [preprint]

Demystifying Border of Structured Algebraic Circuits via Differential Equations
Meghana Bhat, Pranjal Dutta, Nimrod Kaplan, Amir Shpilka, **Devansh Shringi**
Submitted manuscript, 2026 [manuscript]

Reconstruction of Depth-3 Arithmetic Circuits with Constant Top Fan-In
Shubhangi Saraf, **Devansh Shringi**, Narmada Varadarajan
STOC 2026 - Symposium on Theory of Computing [paper]

Reconstruction of Depth 3 Arithmetic Circuits with Top Fan-In 3
Shubhangi Saraf, **Devansh Shringi**
CCC 2025 - Computational Complexity Conference [paper]

Faster & Deterministic FPT Algorithm for Worst-Case Tensor Decomposition
Vishwas Bhargava, **Devansh Shringi**
ICALP 2025 - International Colloquium on Automata, Languages, and Programming [paper]

On the Multilinear Complexity of Associative Algebras
Markus Bläser, Hendrik Mayer, **Devansh Shringi**
STACS 2023 - Symposium on Theoretical Aspects of Computer Science [paper]

Explicit construction of $q+1$ regular local Ramanujan graphs, for all prime-powers q
Rishabh Batra, Nitin Saxena, **Devansh Shringi**
computational complexity, 32, article 2 (2023) [paper]

Selected Talks

STOC 2026 June 2026
Reconstruction of Depth-3 Circuits with Constant Top Fan-In

Workshop on Algebraic Complexity Theory, University of Copenhagen June 2026
Reconstruction of Depth-3 Circuits with Constant Top Fan-In [video]

Theory Seminar, STCS, TIFR May 2026
Reconstruction of Depth-3 Circuits with Constant Top Fan-In [video]

Theory Seminar, University of Toronto Apr. 2026
Reconstruction of Depth-3 Circuits with Constant Top Fan-In

Theory Seminar, IIT Bombay Dec. 2025
Reconstruction of Depth-3 Circuits with Top Fan-In 3

CCC 2025, Fields Institute Aug. 2025
Reconstruction of Depth-3 Circuits with Top Fan-In 3 [slides] [video]

ICALP 2025, Aarhus University July 2025
Faster & Deterministic FPT Algorithm for Worst-Case Tensor Decomposition [slides]

Teaching Experience

Co-instructor, University of Toronto Jan. - Apr. 2026
Algorithm Design, Analysis and Complexity (CSC373)

Teaching Assistant, University of Toronto Sept. 2022 - Present
Graduate courses: Algebraic Gems in TCS; Expanders and Pseudorandom Graphs; Computational Complexity; Algebraic Complexity.
Undergraduate courses: Formal Languages and Automata; Computational Complexity and Computability; Introduction to the Theory of Computation; Algorithm Design, Analysis and Complexity; Algorithmic Game Theory and Mechanism Design.

Teaching Assistant / Tutor, IIT Kanpur Oct. 2020 - Apr. 2022
Probability for Computer Science; Logic; Algorithms II; Fundamentals of Computing.

Awards and Honors

Proficiency Medal, IIT Kanpur 2022
Best master's thesis in Computer Science.

DAAD-WISE Scholarship 2020
Awarded for summer research; research visit cancelled due to COVID-19.

Bronze Medal, International Olympiad on Astronomy and Astrophysics 2016
Represented India.

Reviewing Experience

External conference reviewer: SoCG 2022, ICALP 2024, FOCS 2025, ITCS 2026, STACS 2026, CCC 2026, SODA 2027.

Journal reviewer: SIAM Journal on Computing (SICOMP).

Academic Service

Organizer, Theory Student Seminar, University of Toronto 2024 - 2026
Organized weekly student seminars for the Theory Group.

Coordinator, Association of Computing Activities, IIT Kanpur 2019 - 2020
Organized departmental events connecting students and faculty.